3.8. In an investigation the isotope ratio $^{18}O/^{16}O$ was found to be 2.045×10^{-3} for fresh water and 2.127×10^{-3} for carbon dioxide in the atmosphere. Calculate the equilibrium constant (mole fractions!) for the reaction

$$H_2{}^{18}O(l) + CO_2(g) = H_2O(l) + C\,{}^{18}O^{16}O(g)$$

3.9. How many ideal stages in an ordinary cascade are required at an $\alpha = 6$ to produce water in which 10% of the hydrogen is deuterium?

3.10. In a distillation column with total reflux, ^{10}B is enriched through exchange distillation of $BF_3O(C_2H_5)_2$ from the natural value of 20 atom % to a product containing 95% ^{10}B. The packed column has a length of 5 m and a diameter of 3 cm. What is the approximate height of a theoretical stage if the enrichment factor is 0.026?

3.11. A gas centrifuge plant is set up in order to enrich UF_6 of natural isotopic composition in ^{235}U. The centrifuges, which each have a length of 100 cm and a diameter of 20 cm, rotate at 40000 rpm. The gas temperature is 70°C.

(a) Prove that the separation factor α in (3.61) can be approximated by e^{δ} according to (3.75) when the product flow is very small compared to the waste flow, and α is not far from 1.

(b) Using this approximation, what is the theoretical separation factor for one unit?

(c) Assuming that the enrichment factor obtained with the centrifuge is only 70% of the theoretical one, what number of units would be required in series in order to achieve UF_6 with 3% ^{235}U?

3.12. How much separative work is needed in order to produce 1 kg of uranium containing 4% ^{235}U from a feed of natural uranium and leaving 0.25% in the tail?

3.10. LITERATURE

J. M. McCrea, On the Isotopic Chemistry of Carbonates and a Paleotemperature Scale, *J. Chem. Phys.* **18** (1950) 849.

J. Biegeleisen, Isotopes, *Ann. Rev. Nucl. Sci.* **2** (1953) 221.

G. H. Clewlett, Chemical separation of stable isotopes, *Ann. Rev. Nucl. Sci.* **1** (1954) 293.

A. P. Vinogradov, *Bull. Acad. Sci. U.S.S.R.* **3** (1954) 3.

S. Epstein, quoted by A. P. Vinogradov.

T. F. Johns, Isotope separation by multistage methods, *Progr. Nucl. Phys.* (ed. O. R. Frisch) **6** (1957) 1.

J. Kistemacher, J. Biegeleisen, and A. O. C. Nier (eds.), *Proceedings of the International Symposium on Isotope Separation*, North-Holland, Amsterdam, 1958.

J. H. Beynon, *Mass Spectrometry and its application to Organic Chemistry*, Elsevier, Amsterdam, 1960.

L. Melander, *Isotope Effects on Reaction Rates*, Ronald Press, New York, 1960.

R. E. Weston Jr., Isotope effects in chemical reactions, *Ann. Rev. Nucl. Sci.* **11** (1961) 439.

H. London, *Separation of Isotopes*, G. Newnes, London, 1961.

A. E. Brodsky, *Isotopenchemie*, Akademie-Verlag, Berlin, 1961.

S. S. Roginski, *Theoretische Grundlagen der Isotopenchemie*, VEB Deutscher Verlag der Wissenschaften, Berlin, 1962.

IUPAC and IAEA, *Isotope Mass Effects in Chemistry and Biology*, Butterworths, London, 1962.

M. L. Smith (ed.), *Electromagnetically Enriched Isotopes and Mass Spectrometry*, Butterworths Sci. Publ., London, 1965.

F. A. White, *Mass Spectrometry in Science and Technology*, J. Wiley, 1968.

R. W. Levin, Conversion and enrichment in the nuclear fuel cycle, in D. M. Elliot and L E. S. Villani, *Uranium Enrichment*, Springer Verlag, 1979.

M. Benedict, T. Pigford, and H. W. Levi, *Nuclear Chemical Engineering*, 2nd ed., McGraw-Hill, New York, 1981.

FAO and IAEA, *Stable Isotopes in Plant Nutrition, Soil Fertility, and Environmental Studies*, IAEA, Vienna, 1991.

J. Andrasko, I. Kopp, Å. Åbrink and T. Skiöld, *J. Forensic Sci*. **38** (1993) 1161.

A calculator for enrichment of uranium isotope mixtures containing several uranium isotopes can be found on the internet at: http://www.wise-uranium.org/nfcjol.html

CHAPTER 4

Nuclear Mass Stability

Contents

4.1. Patterns of Nuclear Stability 65
4.2. Neutron to Proton Ratio 67
4.3. Mass Defect 69
4.4. Binding Energy 72
4.5. Nuclear Radius 74
4.6. Semiempirical Mass Equation 75
4.7. Valley of β-Stability 77
4.8. The Missing Elements: $_{43}$Tc and $_{61}$Pm 79
4.8.1. Promethium 79
4.8.2. Technetium 80
4.9. Other Modes of Instability 82
4.10. Exercises 82
4.11. Literature 83

4.1. PATTERNS OF NUCLEAR STABILITY

There are today 255 different nuclei which have shown no evidence of radioactive decay and, hence, are said to be stable with respect to radioactive decay. Currently this means that if they have a half life it is greater than $\sim 10^{20}$ years, which is the longest half-life measured to this date. When these nuclei are compared for their constituent nucleons, we find that approximately 60% of them have both an even number of protons and an even number of neutrons (*even-even nuclei*). The remaining 40% are about equally divided between those that have an even number of protons and an odd number of neutrons (*even-odd nuclei*) and those with an odd number of protons and an even number of neutrons (*odd-even nuclei*). There are now only 4 stable nuclei known which have both an odd number of protons and odd number of neutrons (*odd-odd nuclei*); $^{2}_{1}$H, $^{6}_{3}$Li, $^{10}_{5}$B, and $^{14}_{7}$N, because recently $^{50}_{23}$Vwas found to be instable with a very long half-life. It is significant that the first stable odd-odd nuclei are abundant in the very light elements (the low abundance of $^{2}_{1}$H has a special explanation, see Ch. 12). The last nuclide is found in low isotopic abundance (0.25%) and has an extremely long half-life, 1.4×10^{17} y.

Considering this pattern for the stable nuclei, we can conclude that nuclear stability is favoured by even numbers of protons and neutrons. The validity of this statement can be confirmed further by considering for any particular element the number and types of stable isotopes; see Figure 4.1. Elements of even atomic number (i.e. even number

Radiochemistry and Nuclear Chemistry
ISBN 978-0-12-405897-2, http://dx.doi.org/10.1016/B978-0-12-405897-2.00004-5

© 2013 Elsevier Inc.
All rights reserved.